

3

1

2

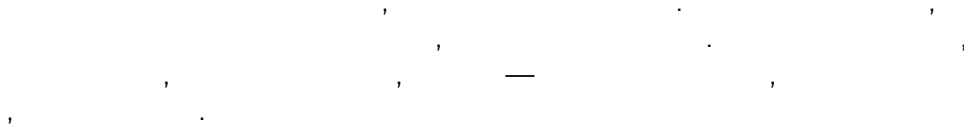
1

(white) (Daisyworld). , - — (black)

2



Daisyworld



Daisyworld

.

,

Daisyworld

Daisyworld

. (). :

— luminosity — $(\frac{L}{L_{\odot}})$ — albedo_black, albedo_white —
 . — surface_albedo — . — max_age — .

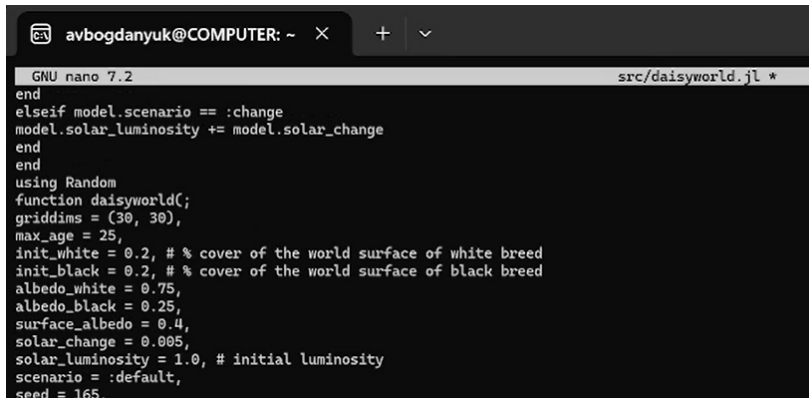
(. 1).

```
[29816b5a] + LibSSH2_jll v1.11.3+1
[14a3606d] + MozillaCACerts_jll v2025.11.4
[458c3c95] + OpenSSL_jll v3.5.4+0
[83775a58] + Zlib_jll v1.3.1+2
[8e850ede] + nghttp2_jll v1.64.0+1
[3f19e933] + p7zip_jll v17.7.0+0
✔ Проект создан: project
♦ Перейдите в директорию: cd project
avboggdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03$ ls
presentation project report setup_project.jl
avboggdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03$ cd project
avboggdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ ls
Manifest.toml Project.toml README.md _research data notebooks papers plots scripts src test
```

1: project

src/daisyworld.jl.

(. 2).



```
avbogdanyuk@COMPUTER: ~ × + ▾
GNU nano 7.2 src/daisyworld.jl *
end
elseif model.scenario == :change
model.solar_luminosity += model.solar_change
end
end
using Random
function daisyworld(;
griddims = (30, 30),
max_age = 25,
init_white = 0.2, # % cover of the world surface of white breed
init_black = 0.2, # % cover of the world surface of black breed
albedo_white = 0.75,
albedo_black = 0.25,
surface_albedo = 0.4,
solar_change = 0.005,
solar_luminosity = 1.0, # initial luminosity
scenario = :default,
seed = 165,
```

2: src/daisyworld.jl

scripts/daisyworld.jl (3).

```
avbogdanyuk@COMPUTER: ~  +  v
GNU nano 7.2 scripts/daisyworld.jl *
using DrWatson
@quickactivate "project"
using Agents
using DataFrames
using Plots
include(scrdir("daisyworld.jl"))
using CairoMakie
model = daisyworld()
daisycolor(a::Daisy) = a.breed
plotkwargs = (
  agent_color=daisycolor, agent_size = 20, agent_marker = '☆',
  heatarray = :temperature,
  heatkwargs = (colrange = (-20, 60),),
)
plt1, _ = abmplot(model; plotkwargs...)
step!(model, 5)
plt2, _ = abmplot(model; heatarray = model.temperature, plotkwargs...)
step!(model, 40)
plt3, _ = abmplot(model; heatarray = model.temperature, |
```

3: scripts/daisyworld.jl

tangle.jl scripts/daisyworld.jl,
(. 4).

```
avbogdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ nano s
avbogdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulation-modeling/labs/lab03/project$ julia
/tangle.jl scripts/daisyworld.jl
```

4:

7

,

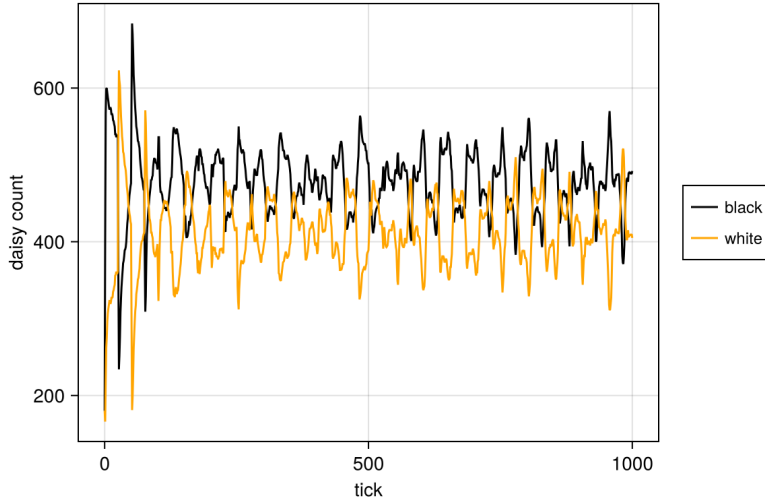
(. 5).

Daisyworld,

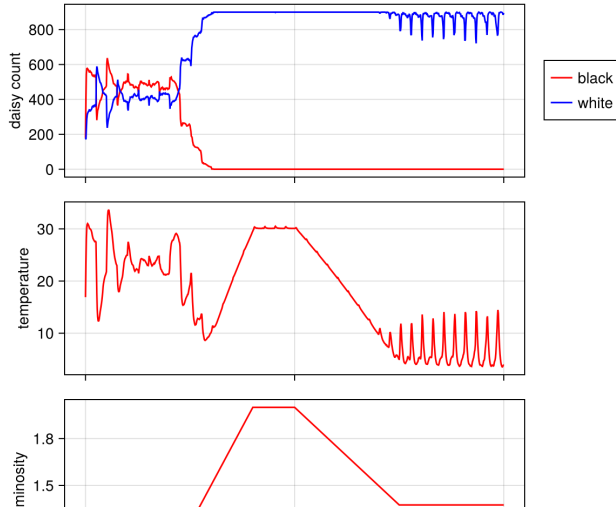
```
avbogdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulatio
n-modeling/labs/lab03/project$ ls scripts
daisyworld          daisyworld-count__param.jl      daisyworld.jl
daisyworld-animate.jl  daisyworld-luminosity          daisyworld__param
daisyworld-count      daisyworld-luminosity.jl       daisyworld__param.jl
daisyworld-count.jl   daisyworld-luminosity__param   intro.jl
daisyworld-count__param  daisyworld-luminosity__param.jl  tangle.jl
avbogdanyuk@COMPUTER:~/work/study/2026-1/2026-1==study--simulation-modeling/2026-1--study--simulatio
n-modeling/labs/lab03/project$ |
```

5:

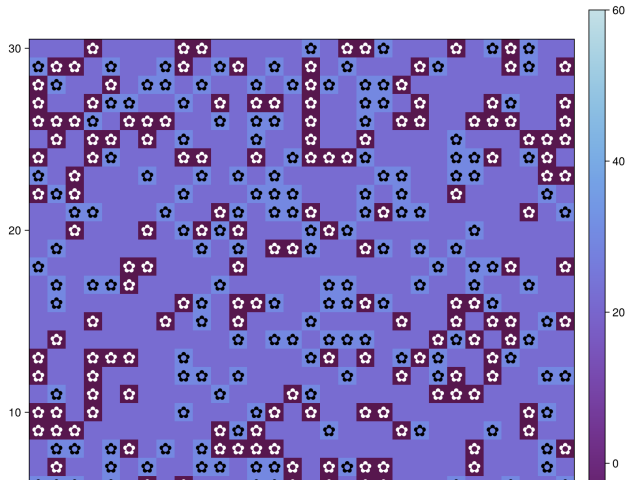
(. 6).



(. 7).



(. 8).



(white) (Daisyworld). , - — (black)

- ① . Watson A. J., Lovelock J. E. Biological homeostasis of the global environment: the parable of Daisyworld // Tellus B: Chemical and Physical Meteorology. — 1983. — Jan. — Vol. 35, no. 4. — P. 284. — ISSN 0280-6509. — DOI: 10.3402/tellusb.v35i4.14616.